

Assignment 1 – Internet Explorer

Client–Server Architecture and What Happens When You Open a Website

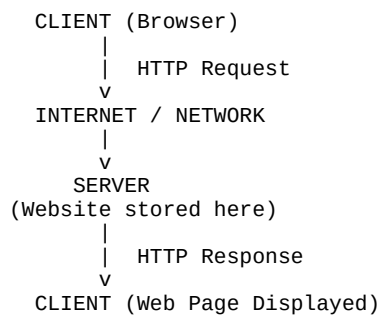
1. Client–Server Architecture

Client–Server Architecture is a network model where a client (user device) sends a request to a server, and the server processes the request and sends back a response. This model is widely used on the internet for websites and applications.

Main Components:

- 1 **Client:** The device used by the user such as a laptop, mobile phone, or tablet. Web browsers like Chrome, Edge, and Firefox act as clients.
- 2 **Server:** A powerful computer that stores websites, applications, and databases.
- 3 **Internet / Network:** The communication medium that connects clients and servers.
- 4 **Request and Response:** The client sends an HTTP request and the server returns an HTTP response.

Simple Client–Server Diagram:



2. What Happens When You Open a Website

- 1 **Step 1: User enters URL** – The user types a website address such as <https://www.google.com> in the browser.
- 2 **Step 2: DNS Lookup** – The browser contacts the Domain Name System (DNS) to convert the domain name into an IP address.
- 3 **Step 3: Request Sent to Server** – The browser sends an HTTP/HTTPS request to the server.
- 4 **Step 4: Server Processes the Request** – The server retrieves website files such as HTML, CSS, JavaScript, and images.
- 5 **Step 5: Server Sends Response** – The server sends the webpage data back to the client.

6 Step 6: Browser Displays Web Page – The browser renders the page and displays it to the user.

Summary: Client–Server architecture allows communication between users and web servers. The browser sends requests, servers process them, and webpages are returned and displayed.